AutoDESN: standardized readout and RHS sensitivity

C1 supplemental report | completed numerical evidence | 23 October 2026

The earlier AutoDESN comparison standardized the response using observations 1:500 and divided readout features by sqrt(d). It did not z-score each readout column. This study adds frozen per-column standardization using the same helpers as Q-DESN and reruns the requested normalized RHS prior.

Setting	Requested fit
Readout features	Mean 0 and sample variance 1 on rows 501:2500; frozen thereafter
Prior controls	tau0 = 0.001; slab_scale = 1; slab_df = 4
Hierarchy	u and local scales half-Cauchy(0,1); tau = tau0 u; c^2 ~ inverse-gamma(2,2)
Other priors	Intercept N(0,10); observation variance inverse-gamma(3,1)
Protocol	500 washout; 2,000 training; 500 held out; 471 origins x 30 leads
Forecast budget	192 joint posterior draws x 8 recursive innovation paths; posterior fit once

Requested fit pooled RMSE: **0.628700** (-3.08% relative to the earlier RHS run). This is a descriptive result on the same previously examined series, not an independent replication or a test of general superiority.

All reported standardized fits passed the declared chain checks and the independent rank-normalized R-hat / bulk and tail ESS screen. Shorter failed attempts are retained and counted in inference cost. Exact diagnostics and their limits appear below.

The historical prefix-scale sampler failed architecture exploration even at 64,000 retained draws per chain; it issued no forecasts. The revised sampler uses layerwise auxiliary slots and an exact product-preserving global/local update. Both preserve the normalized RHS marginal target. This change was made using training-chain diagnostics only.

The first layerwise proposal also failed the declared between-chain architecture-frequency screen. Its training occupancy was used once to freeze a new proposal: 20% original architecture prior plus 80% training occupancy. Fresh production chains then passed. Every architecture retains positive proposal probability, and the exact Hastings ratio preserves the same posterior. No held-out scores entered this change.

This is an exploratory sensitivity analysis. No further tau0/slab search used held-out errors. Overlapping rolling-origin pairs are dependent. Native AutoGP and original-basis outputs are reused, verified references; their inference is not represented as a new run.

Pooled predictive scores

Every row contains exactly 14,130 origin-horizon pairs. RMSE, MAE, CRPS and negative log predictive density (NLL) are in original response units or the corresponding log-density scale. Smaller is better for these four scores.

Model	RMSE	MAE	CRPS	NLL
RHS standardized: 0.001 / 1	0.628700	0.446741	0.339121	0.953479
RHS standardized: 1 / 2	0.628849	0.445357	0.339030	0.953214
Ridge standardized	0.627413	0.445001	0.338207	0.949743
Ridge original	0.647616	0.463250	0.350056	0.981805
RHS original: 1 / 2	0.648673	0.464382	0.350757	0.983525
Native AutoGP	0.617735	0.432304	0.340371	0.977661

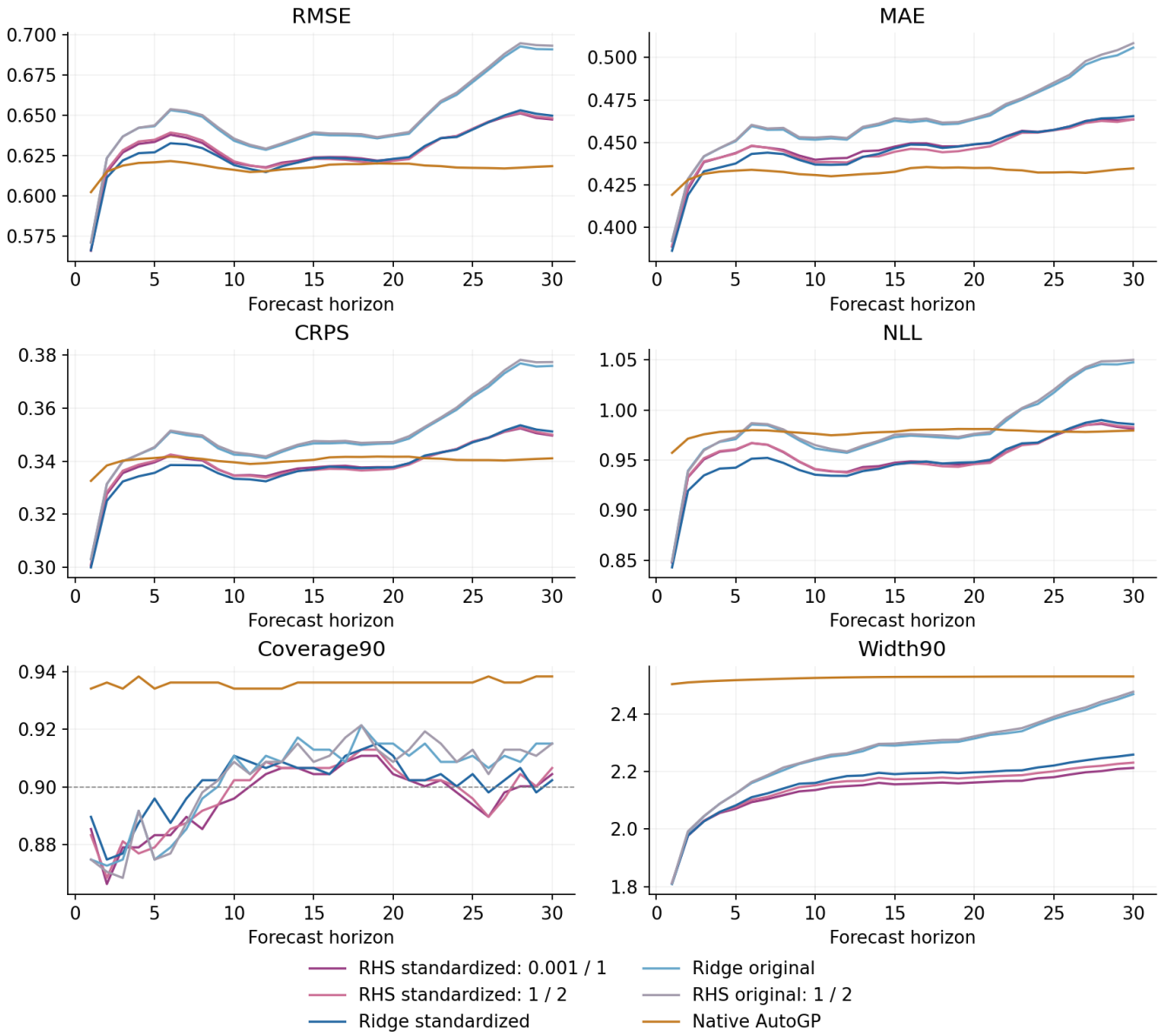
Predictive intervals

Model	Cov. 50%	Width 50%	Cov. 90%	Width 90%
RHS standardized: 0.001 / 1	0.6451	0.8746	0.8964	2.1320
RHS standardized: 1 / 2	0.6515	0.8807	0.8976	2.1462
Ridge standardized	0.6544	0.8863	0.9012	2.1610
Ridge original	0.6536	0.9286	0.9033	2.2621
RHS original: 1 / 2	0.6524	0.9309	0.9032	2.2684
Native AutoGP	0.7129	1.0339	0.9361	2.5259

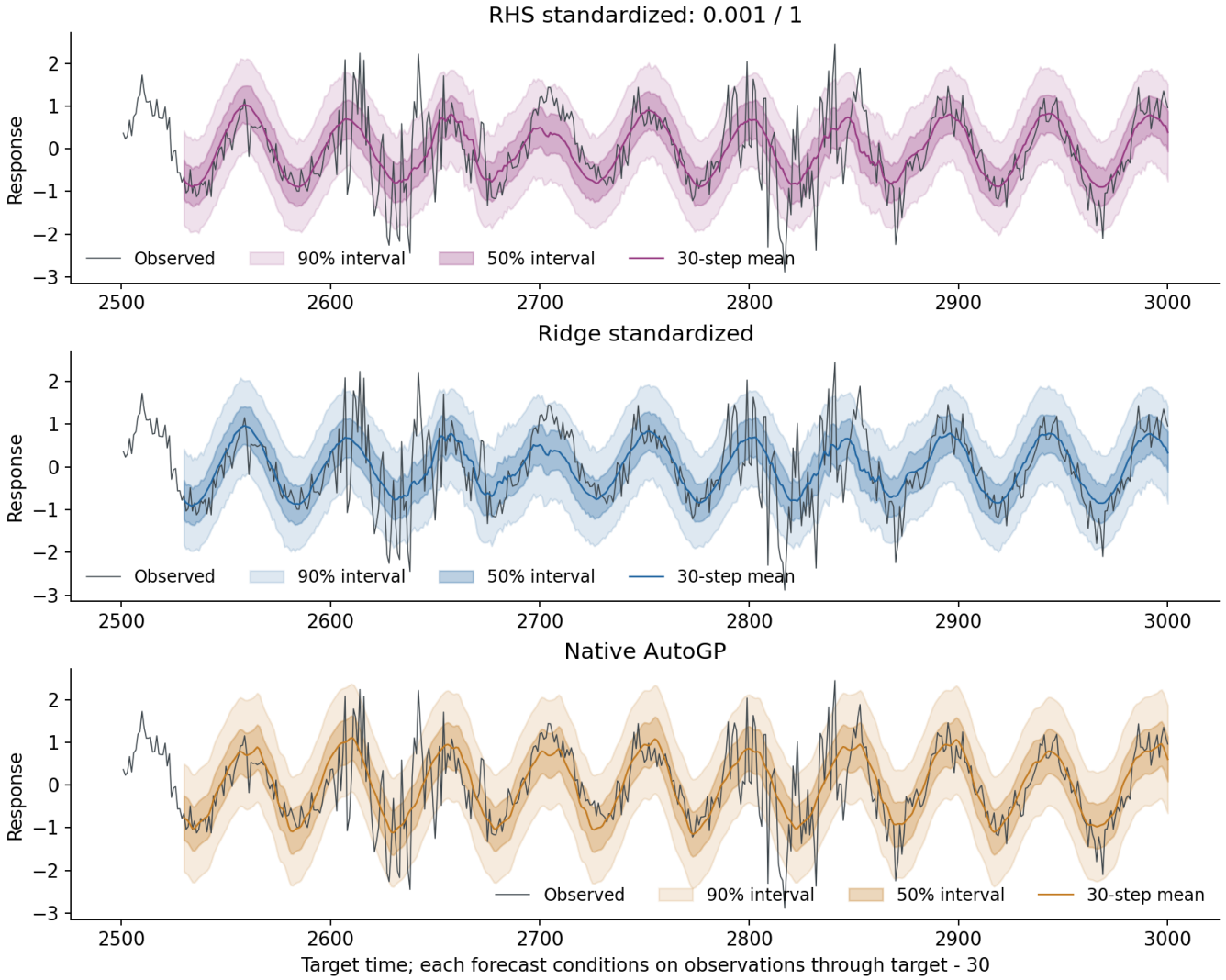
Intervals and CRPS are computed from recursive predictive draws. NLL uses the conditional Gaussian mixture with simulated past paths; it is subject to finite predictive Monte Carlo error. The budget sensitivity below checks that numerical approximation on three fixed origins.

The scaling-only RHS case retains tau0=1 and slab_scale=2. Standardized ridge uses the same standardized bank as both RHS fits. Standardization changes prior geometry, so the original-basis ridge/RHS references are contextual comparisons rather than a controlled prior-only comparison.

Forecast quality by horizon

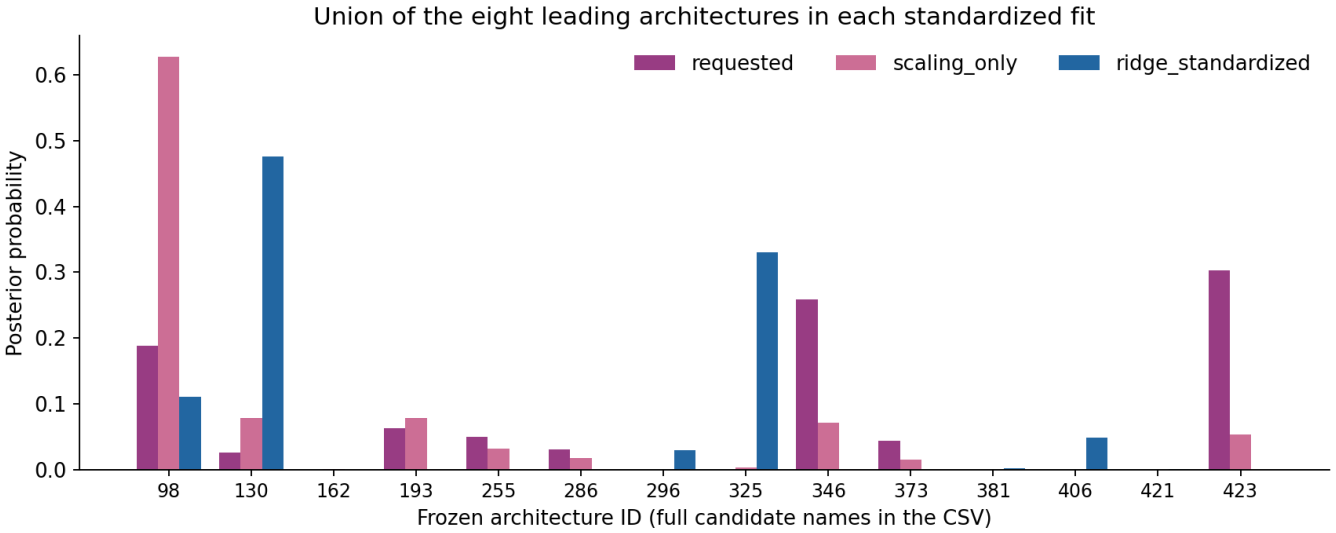


Rolling forecasts and predictive intervals



These panels show horizon 30 at every rolling origin. Observed reservoir states are updated between forecast origins. Inside each issued forecast, future responses are simulated recursively, never read from the held-out series.

Architecture averaging and shrinkage



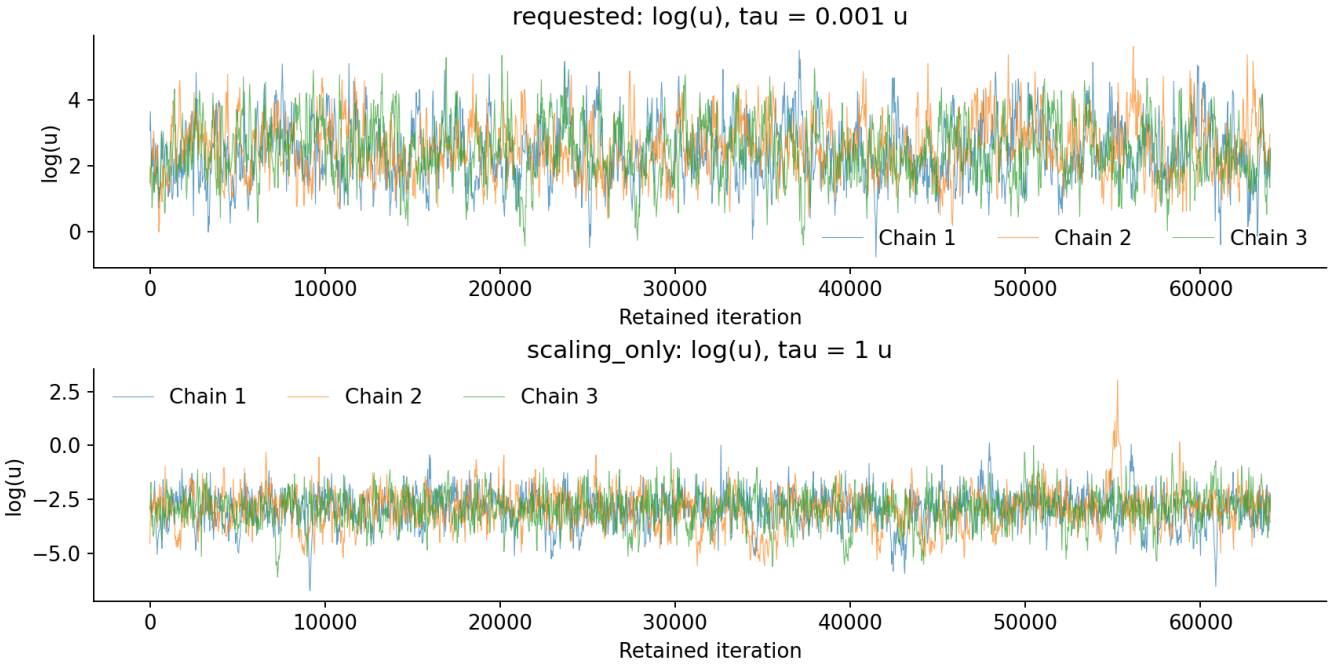
Scenario	Leading model	Mass	tau median [5%,95%]	c^2 median [5%,95%]	Diagonal effective slopes
requested	423: broad_378	0.30236	0.01111 [0.002913, 0.05483]	1.067 [0.3969, 4.761]	41.6 [19.26, 73.16]
scaling_only	98: broad_026	0.62772	0.05476 [0.01255, 0.183]	4.089 [1.503, 18.46]	43.4 [32.79, 76.96]
Ridge standardized	130: broad_058	0.47625	-	-	-

tau0 is a prior multiplier, not a fixed posterior value of tau. Both u and c^2 are sampled. The displayed slab values are posterior draws of c^2, not the slab_scale hyperparameter. Local scales remain coefficient-specific; their distribution is not replaced by a common penalty.

The effective-slope diagnostic sums the diagonal-design approximation to 1 minus the shrinkage factor. Correlated reservoir columns prevent interpreting it as an exact degrees-of-freedom count. The CSV also records aggregate log-local scale traces.

Architecture probabilities are posterior occupancy averages across three chains. Proposals retain positive mass on every architecture and include the exact Hastings correction. No test-error screening replaces Bayesian architecture averaging.

MCMC convergence and numerical validation



Scenario	Max rank R-hat	Min bulk ESS	Min tail ESS	Retained / chain
requested	1.00539	816.5	1394.2	64000
scaling_only	1.00267	940.8	974.5	64000
ridge_standardized	1.00269	1259.5	2493.3	6000

The independent screen includes continuous parameters and model indicators with posterior probability >=1%. Continuous rank-normalized split/folded R-hat must be <=1.01, bulk/tail ESS >=400. Binary model indicators use rank R-hat and frequency ESS; that ESS is shown in both ESS columns because a binary tail quantile can be constant. The R screen also checks spectral ESS and between-chain architecture frequency differences. These diagnostics provide evidence for monitored quantities; they do not prove convergence or estimate rare-model probabilities precisely.

Requested RHS chains visited 13, 13, 14 distinct architectures and made 15841, 15429, 15640 retained-iteration architecture switches. Maximum between-chain architecture-frequency range: 0.01575. All architectures with posterior mass >=1% passed the architecture R-hat and ESS screen.

Sampling-budget sensitivity

Scenario	Burn / chain	Kept / chain	Max R-hat	Min spectral ESS	Pass
requested	10000	64000	1.00539	974.1	True
scaling_only	2000	6000	1.04318	176.9	False
scaling_only	4000	16000	1.01879	454.6	False
scaling_only	6000	32000	1.00360	711.4	False
scaling_only	10000	64000	1.00241	1465.4	True
ridge_standardized	2000	6000	1.00269	1210.8	True
legacy_prefix_requested	2000	6000	1.08538	63.6	False
legacy_prefix_requested	4000	16000	1.04443	188.0	False
legacy_prefix_requested	6000	32000	1.02775	345.2	False
legacy_prefix_requested	10000	64000	1.00808	705.8	False
requested_initial_proposal	2000	6000	1.04901	104.4	False
requested_initial_proposal	4000	16000	1.02183	280.6	False
requested_initial_proposal	6000	32000	1.00724	493.0	False
requested_initial_proposal	10000	64000	1.00224	1001.2	False

The displayed R-hat and spectral ESS columns concern continuous parameters. Pass additionally requires the architecture R-hat/ESS and absolute chain-frequency range gates.

Predictive Monte Carlo sensitivity

Scenario	N x K	RMSE (3 origins)	CRPS	NLL
requested	192 x 8	0.32632	0.21196	0.60364
requested	384 x 32	0.32497	0.21112	0.60401
scaling_only	192 x 8	0.33115	0.21377	0.60856
scaling_only	384 x 32	0.33051	0.21348	0.61013
ridge_standardized	192 x 8	0.34757	0.22071	0.62745
ridge_standardized	384 x 32	0.34681	0.21971	0.62727

The full CSV contains all 24 settings per model: origins 2500, 2735, 2970; N=192/384; K=4/8/16/32. These reuse one qualified posterior per model. The 90 scored pairs in a setting are dependent and are not the full 14,130-pair comparison.

The two standardized RHS priors differ in full-comparison RMSE by about 0.00015. Predictive-budget changes on these three origins produce larger shifts. This sensitivity does not support ranking the two RHS settings at that precision.

Measured runtime and reproducibility

Shared raw reservoir construction: 145.45 s wall time (180.61 s summed candidate compute). Frozen readout preparation, including column standardization and sufficient statistics: 18.12 s. Bank loading, compilation and reporting are not inference.

Scenario	Pilots (s)	Inference incl. failed attempts (s)	Forecast compute (s)	State update (s)	Parent peak RSS (GiB)
requested	0.00	690.73	90.16	2.99	4.89
scaling_only	23.27	1085.36	114.27	1.19	NA
ridge_standardized	0.00	81.02	146.08	2.78	4.25

Forecast compute is the sum of per-origin computation, not parallel wall time. State preparation, scoring, exports and predictive-budget sensitivity are excluded from that sum. Parent VmHWM excludes forked workers and is not a unique-memory estimate for the entire process tree. Cgroup observations include file cache and concurrent dependency setup.

The initial scaling-only RSS sample used a namespace-local PID against host /proc and is invalid; it is reported as NA. The corrected observer resolves the process namespace explicitly. Ridge forecasting was interrupted after 98 origins and resumed from its saved fit and per-origin checkpoints; its measured peak covers the observed resume invocations, including a failed fork attempt. That failed resume took 255.02 s wall time, separate from completed forecast compute. No refit was needed.

Additional failed prefix-sampler development cost: 1239.80 s inference plus 28.08 s pilot computation. These produced no forecasts and are retained in the budget/diagnostic evidence.

The requested fit additionally used 1354.04 s of completed training-chain computation with the initial layerwise proposal, plus 25.44 s of pilot computation. These failed attempts supplied the frozen proposal and produced no forecasts. They are additional to the qualified-fit inference row, not a new prior or a selection on forecast errors. Interrupted partial work is recorded in invocation logs and is excluded from completed-chain compute totals.

Reused Native AutoGP reference: two runs with eight SMC particles each; summed training time 1470.06 s; original native run/export time 1522.27 s. Its later rolling conditioning/scoring/export replay took 22.29 s with peak RSS 0.237 GiB. A pure GP forecasting timer and native-training peak memory were unavailable. These are different timer scopes and cannot support a clean end-to-end speedup claim.

A separate short-chain computational benchmark (200 rows, four standardized designs, up to 64 slopes, 100 burn +800 retained draws; three timing repetitions) took median 1.414 s in R and 0.864 s with the native kernels, a 1.64x ratio. Coupled coefficient/scale draws differed by at most 4.96e-12, with identical model paths. Setup and compilation were excluded. These short runs measure computation, not convergence.

Audit details

All 428 admitted architectures have the same raw reservoirs and candidate identities in all three new cases. Maximum absolute training feature mean: 7.09e-14; maximum sample-SD error: 2.22e-16. No constant columns were found in this bank. The API freezes a constant-column exclusion mask when necessary.

Inference conditions on these frozen training transformations. Uncertainty in the estimated feature means and scales is not jointly inferred. The intercept is unscaled and has a proper prior; transport to raw reservoir coordinates includes its centering correction.

Validation covers shared Q-DESN scaling parity, frozen transforms, intercept transport, direct/batched recursive forecast parity, backward-compatible ridge draws, normalized RHS densities, Gaussian integrated likelihoods, scale updates, architecture ratios, and R/C++ agreement. Saved path snapshots independently reproduce scores at the first, middle and last origins. See validation_status.json and the archived test logs for actual pass/skip counts.

Current installed regression assertions: 639 passed, zero failures/errors/warnings, one loaded-DLL context skip covered separately by the source-only checks. The 32 independent small-posterior reference checks passed. Prior-preserving interweaving has independent conditional quadrature and no-information prior-CDF checks; the historical default sampler reproduces the earlier draws exactly.

Release status remains draft. The earlier scoped R CMD check recorded one inherited nonportable-path error, three existing documentation/non-ASCII warnings and two notes. This update installs and passes the focused checks; a clean repository-wide release check is not claimed.

Reproduce from feature/autodesn-rhs-20261022 using README.md: build the shared bank, run the initial training attempts with run.R, then run_rebalanced.R for the requested production fit. Run scaling_only and ridge_standardized sequentially, forecast_serial.R as needed, sensitivity.R, export.R, archive.R, replay.R, analysis.py and report.py. Configuration and saved contracts fix budgets, seeds and source hashes. Restore the numeric evidence for analysis without recomputing inference.

Code and draft PR: https://github.com/AntonioAPDL/exdqlm/pull/3. Original report and reproducibility evidence remain under validation/autodesn/rhs_comparison/evidence/. This supplement preserves those results.